Photographing a meteor shower

Print it, fold it, take it out with you.

Fragments of the Universe
bhapstar.com

The whole thing in one paragraph

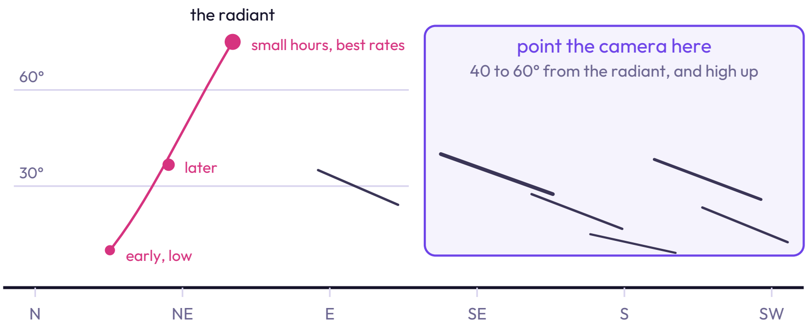
You do not chase meteors. A meteor lasts a fraction of a second, so the only thing that decides whether you catch one is how much of the night your shutter was open. Point at a good patch of sky, set the camera or the phone running, and take hundreds of frames. Every one of them is also a Milky Way exposure, so a quiet night is never a wasted one.

THE SHOWERS WORTH PLANNING AROUND

Shower	Peak	Rough rate at a dark site	Worth knowing
Quadrantids	3 to 4 January	25 to 50 an hour	The peak is very short, often only a few hours. Get the night right or miss it entirely
Lyrids	22 to 23 April	10 to 20 an hour	Modest, but it breaks the long gap between January and August
Eta Aquariids	5 to 6 May	20 to 40 an hour	Dust from Halley's Comet. Fast, and best in the hours before dawn
Perseids	12 to 13 August	50 to 80 an hour	Warm nights and a high radiant make this the easiest one to sit through
Orionids	21 to 22 October	10 to 20 an hour	Halley's Comet again, from the other side of its orbit
Leonids	17 to 18 November	10 to 15 an hour	Quiet most years, with occasional storms decades apart
Geminids	13 to 14 December	60 to 100 an hour	The best of the year. Bright, slow, and coming from an asteroid rather than a comet
Ursids	22 to 23 December	5 to 10 an hour	Small, and easy to combine with the Geminids the week before

Those rates are what a patient person sees by eye at a genuinely dark site, with the radiant high and the moon out of the way. Published figures are often higher because they describe an idealised sky. Your camera will record fewer than your eyes do, because it only looks at one part of the sky at a time.

POINT AWAY FROM THE RADIANT



Meteors near the radiant are travelling towards you and appear as short stubs. The long, photogenic ones are always well off to the side. Anywhere high and dark works, so use that freedom to pick the direction with the least streetlight and the best foreground.

CHECK THE MOON FIRST

This is the step that decides the night. A bright moon washes the sky out exactly the way a city does, and hides most of the fainter meteors. A shower peaking two days after full moon is worth skipping. A shower on a bright night is often still worth shooting in the window after the moon has set, even if that window starts at three in the morning.

STAY LATE

Rates almost always build through the night. Early on the radiant is still low and much of the shower is happening below the horizon. By the small hours the Earth has turned to face into the stream. If you can only be out for part of the night, arrive late rather than leaving early.

WHAT TO SET

Camera
Manual, RAW, lowest f-number, ISO 3200 at a dark site, 15 to 20 seconds on a wide lens. Lock the intervalometer on.

Phone
Night mode on a tripod, or Pro mode at ISO 3200 and 20 seconds, fired again and again.

Either
Shorten any long night mode. A meteor in a four minute frame is swamped by four minutes of skyglow; the same meteor in a 20 second frame stands out sharply.

ON THE NIGHT, AND AFTERWARDS

Take more frames than feels sensible

One exposure is a lottery ticket. Thirty is a fair chance. A few hundred, across two or three hours, is how people come home with the shot. Expect most frames to hold nothing, which is completely normal.

What decides the night

More power than you think, because cold drains batteries fast. Card space: three hours of 20 second frames is around 500 files. Check the front of the lens every half hour for dew. Dress for two hours longer than you plan to stay.

Three pictures after

Pick the best single meteor frame and process it alone. Combine several meteors onto one sky, which is fine as long as you say so. Average or stack the whole run for a clean Milky Way or a star trail.

The full guide, with examples

Full walkthrough with example images, the radiant explained properly, and the separate phone and camera cards with the complete settings for each.

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